

360RVW: Fusing Real 360° Videos and Interactive Virtual Worlds

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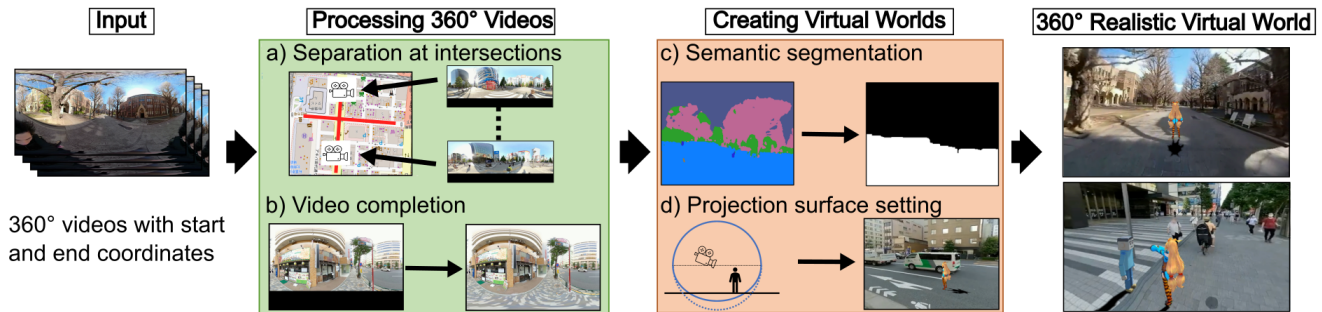


Figure 1: We propose a system to create 360° realistic virtual worlds (360RVW) from omnidirectional street videos that allow scene exploration and user interactions.

ABSTRACT

We propose a system to generate 360° realistic virtual worlds (360RVW) for the interactive spatial exploration of omnidirectional street-view videos. Our 360RVW enables users to explore photorealistic scenes with digital avatars, and interact with others. To create the virtual worlds our system only requires 360° videos with annotations of the start and end camera coordinate as input. We first detect street intersections to divide the input videos and remove the camera operator from the recordings using a video completion technique. Next, we analyze the 3D structure of the scene using semantic segmentation to define walkable areas. Finally, we render the environment using an ellipsoid projection surface to achieve a more realistic integration of the avatar into real-world 360° videos. The whole process is largely automated, enabling users to produce realistic and interactive virtual worlds without specialized skills or time-consuming manual interventions.

CCS CONCEPTS

• **Information systems** → **Multimedia content creation**; • **Computing methodologies** → *Virtual reality*.

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KEYWORDS

omni-directional video, virtual reality, interface

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1 INTRODUCTION

In recent years, virtual worlds have allowed users to connect with others and explore locations from all over the world. Even large real-world gatherings have successfully been transferred to online events with a Halloween celebration held in a virtual recreation of Shibuya attracting over 400,000 visitors [9]. Unfortunately, creating such virtual worlds is time-consuming and requires specialized skill to design convincing 3D environments. Moreover, there are limits to how well 3D models can reproduce the real world because living things or moving cars don't exist. To address this challenge, we use 360° videos instead of building complex 3D models. Unlike 3D models, 360° videos can easily be created due to the widespread availability of 360° cameras and convey a realistic impression of the environment, however, they are generally non-interactive and do not allow free spatial exploration. Therefore, in this paper, we seek to generate photorealistic and interactive low-cost virtual worlds by fusing virtual 3D environments and 360° videos.

To avoid the manual creation of 3D worlds, many systems using 360° media have been proposed. In Google Street View (GSV) [5],

360° photos are used to give users an impression of the environment, however, the images are only taken at discrete positions and do not offer interactivity. To improve the experience, Geollery [1] combines images from GSV with social media functions and 3D city models. Considering 360° videos, Movie Map [8] proposed a system to create a connected world from 360° street-view videos enabling users to navigate the world by choosing a walking direction at intersections. Other works focus on how to incorporate user interactions in 360° videos considering video streaming [6] or shared viewing experiences [3]. Furthermore, Tourgether360 [4] proposes a collaborative interface that displays the camera trajectory and the eye of users' avatars to indicate their viewing direction in a 360° video landscape, increasing the spatial awareness and social presence of users.

In this paper, we enrich omnidirectional videos to create 360° realistic virtual worlds (360RVW) by adding free spatial exploration via digital avatars. In contrast to previous works, users can move their avatars in a 3D representation of the real-world video which includes walking people and moving cars and interact directly with others. Furthermore, our system only requires 360° videos and the start and end camera coordinate as input to afterwards automatically generate virtual environments without manual intervention.

2 SYSTEM

Our system enables the automatic generation of virtual environments and only requires 360° videos with defined start and end coordinates as input (Fig. 1).

2.1 Processing the 360° Videos

To separate the input videos for each road segment, we process them following Movie Map [8]. First, the relative camera positions and rotations are estimated with OpenVSLAM [10] in order to assign camera coordinates to each frame based on the start and end coordinates of the videos (Fig. 1 (a)). Next, the coordinates and visual features are used to detect pairs of video frames corresponding to each intersection seen in the videos. Finally, we divide the videos by the intersection frames. This approach enables users to freely explore the virtual world rendered from the videos by changing their direction of movement at road intersections.

In Movie Map [8] the camera operator is excluded from the video by blacking out the ground area. As we need to include a representation of the ground to enable the navigation via avatars, we add an additional processing step by applying the video completion model FGCv [2] before dividing the videos. This module can remove the camera operator from the videos to ensure a natural representation of the floor in the resulting virtual environment (Fig. 1 (b)).

2.2 Creating Virtual Worlds from 360° Videos

Our system uses two main functions to generate a virtual environment from 360° videos. First, we apply semantic segmentation using Orhans' model [7] to obtain labels for each pixel indicating the position of roads and buildings. Based on this, we can construct a binary map that encodes which areas constitute the ground of the scene (Fig. 1 (c)). This enables us to determine whether or not avatars are on the road by projecting the position of the avatar to the coordinate in the binary map. Second, to naturally integrate the

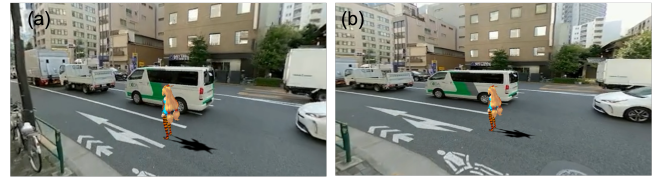


Figure 2: The effect of the projection: (a) In a spherical projection, the avatar seems to float. (b) In the proposed elliptical projection, the avatar appears as a part of the scene.



Figure 3: Applications: (a) Exploring famous tourist attractions. (b) Real-time interaction with other users' avatars.

avatar into the scene, our system changes the projection from the regular sphere projection to a shallow-bottomed ellipse (Fig. 1(d)). In addition, we set the ground plane to the position corresponding to the road in the scene. With these improvements, as shown in Fig. 2, the avatar appears as if it were in a real 3D scene while a spherical projection results in the avatar appearing to float.

2.3 System Implementation

Our system handles the processing of 360° videos and generation of the virtual environment as an offline preprocessing step. To display the scenes, we use Unity [11] and display the environments and avatars in real time. Here, the videos are played back according to the forward movement speed of the avatar allowing the user to experience the scene as if the avatar is walking through the video. Moreover, since our system allows users to explore the videos at the same time, they can not only interact with the environment but also with each other. Overall, users are able to experience the collaborative spatial exploration of 360° videos. In our demo, we will showcase locations of Japan including famous sightseeing spots and busy shopping districts.

3 CONCLUSION

In this work, we propose a system to generate new virtual environments from 360° videos that have both a realistic appearance and interactive content. Our system extends 360° videos to realistic virtual worlds by applying segmentation to define explorable areas and uses video completion models as well as a shallow-bottomed ellipse projection to convincingly render avatars in the scene. As shown in Fig. 3, users can freely enjoy viewing and navigating real locations and interact with other users' avatars. We believe that our system will contribute to the development of realistic and immersive multi-user interactive virtual worlds, which can accelerate the development of the metaverse as an emergent online framework.

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